'Assignment - 03'

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Question-1

(a)

Given that, pic-x can handle 20 different interrupt requests at a certain time. That means,

IRR that track pending interrupts requires 20 bits.
ISR that tracks interrupt currently being served requires 20 bits
IMR that is used to mask specific interrupt lines also requires 20 bits.

(b)

at t= 32 ns, IR6 recieved interrupt and start serving it. It needs 50 ns to serve IR6.

at t= 45 ns, IR5 received interrupt. Since IR5 has higher priority than IR6 in fixed priority mode,

IR6 stops and remaining time = 50 - (45-32) = 37 ns
IR5 starts execution. It needs 20 ns to serve IR5.

at t=52 ns, IR7 and IR0 received interrupt. IR0 has higher priority than IR5. so,

IR5 stops and remaining time = 20 - (52-45) = 13 ns
IR0 starts execution. It needs 37 ns to get served.

at $t = 89$ ns, IR0 finishes and since fixed priority mode, next IR5 starts executing.

at $t = 102$ ns, IR5 finishes and next in line is IR6.

finally, at $t = 139$ ns IR6 finishes.

So, at $t = 139$ ns, interrupt at IR6 is served completely.

(c)

at 32 ns, IR6 is being served

IRR : 0000 0000 0000 0000 0000

ISR : 0000 0000 0000 0100 0000

IMR : 0000 0000 0000 0000 0000

at 45 ns, IR5 arrived and acknowledged

IRR: 0000 0000 0000 0000 0000

ISR : 0000 0000 0000 0110 0000

IMR: 0000 0000 0000 0000 0000

at 52 ns, IR7 and IR0 arrived, IR0 is acknowledged

IRR: 0000 0000 0000 1000 0000

ISR : 0000 0000 0000 0110 0001

IMR : 0000 0000 0000 0000 0000

at 89 ns, IR0 finishes.

IRR: 0000 0000 0000 1000 0000

ISR: 0000 0000 0000 0110 0000

IMR: 0000 0000 0000 0000 0000

at 102 ns, IR5 finishes,

IRR: 0000 0000 0000 1000 0000

ISR: 0000 0000 0000 0100 0000

IMR: 0000 0000 0000 0000 0000

at 139 ns, IR6 finishes and IR7 is acknowledged,

IRR: 0000 0000 0000 0000 0000

ISR: 0000 0000 0000 1000 0000

IMR: 0000 0000 0000 0000 0000

(d)

Upon receiving an INTR, 8086

1) firstly finishes the currently running instruction.

2) checks the interrupt flag. When IF=1, it acknowledges the interrupt by generating two active low pulse through INTA.

3) The first INTA tells PIC to prepare the interrupt vector number and determine highest priority.

4) The second INTA tells pic to place the vector number onto the data bus

5) Then, 8086 reads this vector number, pushes the flags register onto the stack and clear TF and IF.

6) It also pushes cs and IP onto the stack.

7) It calculate the vector address and fetches the new cs and IP from the interrupt table to begin executing ISR.

(e)

8086 clears TF and IF while serving an interrupt.

It clears TF because, when single step debugging mode is off, TF=0, it ensures that the interrupt routine runs without breaking after every instruction.

Again, it clears IF to prevent other maskable interrupt from disrupting the current interrupt, avoiding uncontrolled nesting interrupts.

(f)

Hardware and software interrupts are two types of interrupt in 8086. The difference between them are given below:

Hardware	Software
Hardware interrupts trigger through pins like INTR, NMI	software interrupts are triggered through internal execution of software. for example, INT n
They can occur at any time.	They occur during the execution of the instruction.
INTR can be masked, NMI cannot.	Software interrupts cannot be masked.
IF has influence on hardware interrupts.	IF has no influence on software interrupts.
example; mouse and keyboard interrupt.	example; INT 0 divide-error interrupt.

(g)

There are 65,536 different interrupt types. That means,

$$\log_2(65536) = 16$$

So, 16 data bus lines (16-bits) of PIC-X would be connected to 8086 data bus.

Question-2

(a)

The instruction divides 16-bit value in AX by the 8bit value in BL. The quotient is stored in AL, and the remainder in AH.

The quotient is C95H which is 12 bits. But the AL register is only 8bits. So, a divide error occurs. This generates a Type 0 interrupt.

(b)

Given, physical address = 8B62AH

CS = 8A00H

$\therefore$ IP = 8B62AH − (8A00H × 1011)

= 162AH

for type 0, (0H × 4H) = 00000H

IVT address	Hex Value
00000H	2AH
00001H	16H
00002H	00H
00003H	8AH

(c)

Given, Physical address = 31122H

$\therefore$ IP = 31222H − (3000H × 10H)

= 1122H

Now, since the instruction is 2 bytes long, the next instruction starts at IP = 1124H. So, Physical address of the next instruction is 31224H. first IP will be pushed then IP. So, final values,

Address	HEX values
AFFF4H	24H
AFFF5H	11H
AFFF6H	00H
AFFF7H	30H

(d)

Given that, The system must handle 25 interrupt sources and IR3 and IR7 are faulty. So, only 6 pins are working on each PIC.

Therefore, 5 total PICs are required, where 1 is master pic and other 4's are slaves PIC.

(e)

Given, Master's,

IMR = 00010001

ISR = 10000000

Slave's,

IMR = 10010010

ISR = 00000000

Now, each slave receives mask on IR0, IR1. IR1 is masked. So, all slaves forward only their IR0 to master PIC.

Master's IMR has IR0 and IR4 masked. But IR1 is unmasked and higher priority than IR7 that is being currently served at master PIC. So, master interrupts its own IR7 and accepts the request from slave on IR1.

That means, IR0 of the slave connected to Master IR1 is the interrupt that is served first.

Question-3

(a)

Given, $T_{ON} = 55.5$ ns

from the waveform,

$$T = T_{ON} + T_{off}$$

$$\Rightarrow T = T_{ON} + \frac{T}{3}$$

$$\Rightarrow 3T = 3T_{ON} + T$$

$$\Rightarrow T = \frac{3T_{ON}}{2}$$

$$\therefore T = 83.25 \text{ ns}$$

$$\therefore \text{frequency} = \frac{1}{83.25 \times 10^{-9}}$$

$$= 12.012 \text{ MHz} \quad \text{(Ans.)}$$

(b)

Given instruction, MOV BX, [3001H].

Here, the address is odd and so the microprocessor fetches from an unaligned address. Hence, it requires 2 bus cycle to fetch 16 bit data. In the first bus cycle, it reads from 3001H, which is an odd bank. In the 2nd bus cycle, µP reads

from 3002H which is even bank.

Now, total T-states = $2 \times (4+1)\,T$

$= 10\,T$

$\therefore$ total length = 10×83.25 ns

$= 832.5$ ns.

(Ans.)

(c)

We know, duty cycle = $\frac{T_{ON}}{T_{ON} + T_{off}} \times 100\%$

$= \left(\frac{55.5}{83.25}\right) \times 100\%$

$= 66.67\%$

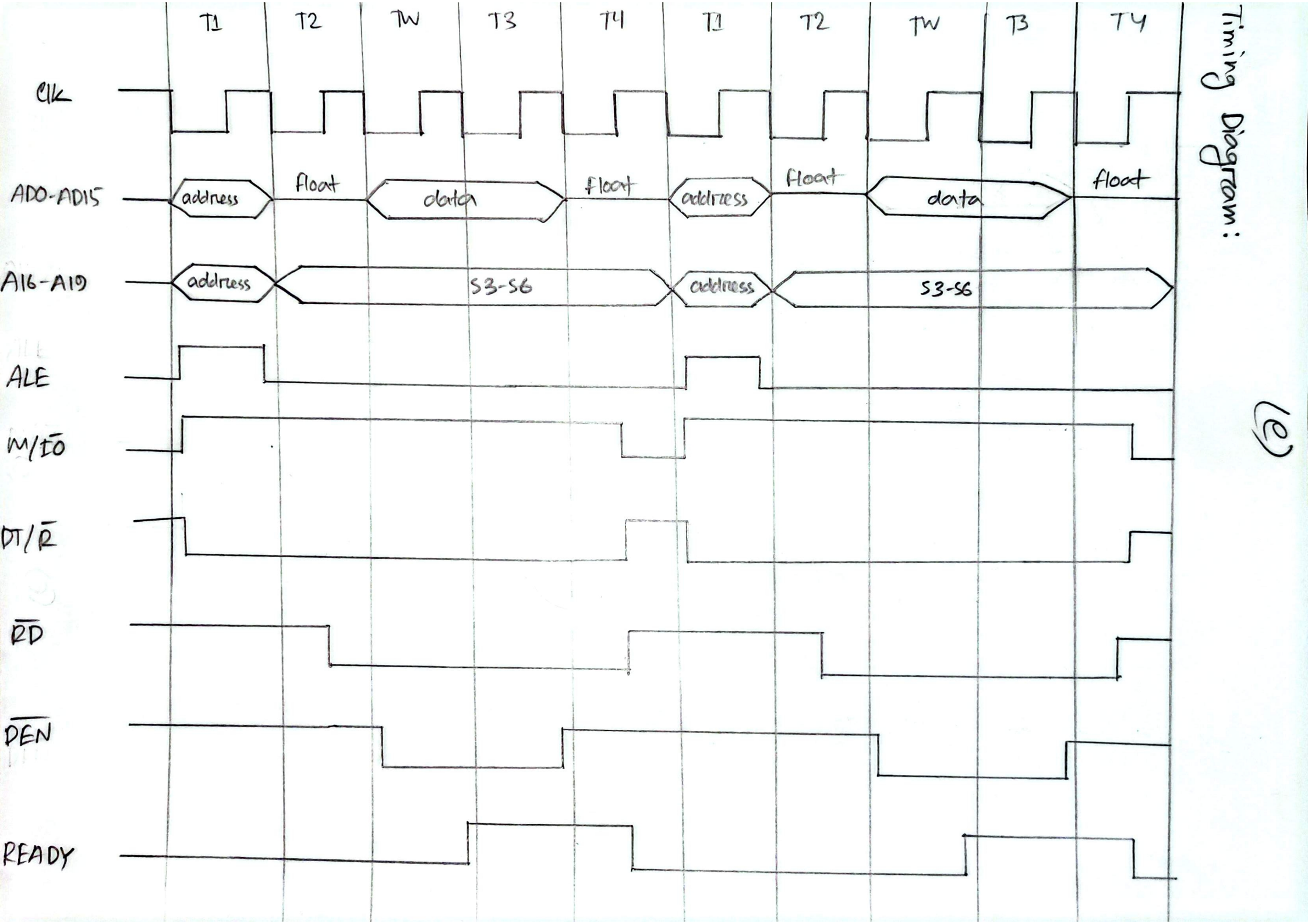
(Ans.)

(d)

We discussed about needing two bus cycle to execute the given instruction.

First Bus Cycle: microprocessor reads from odd bank, data passes through D8-D15. $A_0 = 1$, $\overline{BHE} = 0$.

second Bus Cycle: microprocessor reads from even bank, data passes through D0-D7. $A_0 = 0$, $\overline{BHE} = 1$.



Timing Diagram:

(c)